



Forms Required:

A ____ Contribution from a Recognized Institution

B X Projects that require a human subject

C ____ Same project for a second year

Project
Number

Science Fair Proposal Form

Name of participants: Gabriel Mamane

Grade level: _____

Experimental Project

Project Category:

- Life Sciences
- Health Sciences
- Human Sciences
- Biotechnologies
- Health and Medical Sciences ✓
- Environmental Sciences
- Physical Sciences and Mathematics
- Engineering and Computer Sciences

Title: (30 character maximum)

Night Signal: SpO2 Accuracy

Experimental Question:

How closely do blood-oxygen (SpO2) and heart-rate readings from a low-cost wearable (MAX30102 fingertip sensor + ESP32) agree with a Health Canada-authorized fingertip pulse oximeter, and can software reliably flag oxygen desaturation events (a drop of at least 3 percentage points lasting at least 10 s) in the wearable's recordings?

Preliminary Hypothesis:

If the wearable is worn correctly and only good-quality readings are used, its SpO2 will agree with the reference oximeter within about ± 3 percentage points, because both measure red/infrared light absorption, and the software will flag the dips the reference also shows. This project does not diagnose or confirm sleep apnea.

Dependent variable(s):

Wearable minus reference SpO2 and heart rate (bias, 95% limits of agreement, mean absolute error); number of desaturation events flagged; percentage of recording time with a valid signal.

Independent variable(s):

Measuring device (wearable vs. reference oximeter); test condition (seated rest; optional short supervised breath-holds only if approved by my teacher and the ethics committee).

Controls:

Same participant, finger and cuff tightness; seated and still with warm hands; same room and lighting; paired readings taken at the same moments every 2 minutes; same firmware and detection settings (3-point drop, 10 s minimum, 120 s baseline); same reference oximeter.